CARTESIAN SIMPLICIAL SYSTEMS OF THE DERIVED CATEGORY WITH VANISHING SELF-EXTENSIONS COME FROM CARTESIAN OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two 'we omit the computation' spots (source lines 3207: vanishing of the compositions of the constructed complex; line 3257: commutativity of the squares in the comparison diagram) are auxiliary checks sitting mid-proof, not at the main conclusion, which is fully argued; the comparison towers Y'_{m,n} are the key Postnikov example tensored with K_m; the left adjoints g_{n!} and the component formula used throughout are those established by the restriction-to-components lemma delivered as card L46149 of this batch.

Editorial note. The label lemma-cartesian-object-derived-from-simplicial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The left adjoints g_n! on the simplicial total site and their component formula. Same chapter, \label{lemma-restriction-to-components-site} (card L46149 of this batch). Lines 3187-3194 and 3253-3261: X_n = g_{n!}K_n is formed with the left adjoint of g_n^{-1}, and the Hom computations use g_m^{-1}g_{n!}K_n = the direct sum of f_phi^{-1}K_n over phi : [n] -> [m].
- Derived Categories, existence and comparison of Postnikov systems. Derived Categories chapter, \label{derived-lemma-existence-postnikov-system}. Lines 3224-3229: extends the complex to a Postnikov system when the Hom groups vanish; lines 3296-3301: the second part extends the vertical maps to an isomorphism of Postnikov systems.
- The simplicial resolution of Z on the site. Same chapter, \label{lemma-simplicial-resolution-Z-site}. Lines 3209-3212: the alternating-sum maps mirror this lemma's construction; lines 3275-3278: the bottom-row arrows of the comparison diagram are its maps.

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- The key Postnikov system computing a homotopy colimit. Derived Categories chapter, \label{derived-example-key-postnikov}; Same chapter, \label{lemma-abelian-postnikov}. Lines 3284-3294: the explicit towers Y'_{m,n} = (direct sums of Z)[-n] tensor K_m with hocolim K_m.
- Uniqueness of homotopy colimits in the derived category. Standard property of homotopy colimits in the derived category. Lines 3302-3310: gamma_m : g_m^{-1}K -> K_m is assembled from the compatible gamma_{m,n} by uniqueness of hocolims.

Source reference. Original section 0D8Z: sources/stacks-chow/tags.txt, source lines 16609–16609. Prerequisite pointer invoked at proof line(s) 3268; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D83: sources/stacks-chow/tags.txt, source lines 16578–16578. Prerequisite pointer invoked at proof line(s) 3209; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9I: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3002–3002. Prerequisite pointer invoked at proof line(s) 3267; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WI: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1711–1711. Prerequisite pointer invoked at proof line(s) 3202; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 3176; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE. Let (K_n, K_φ) be a simplicial system of the derived category. Assume*

(1) *(K_n, K_φ) is cartesian,*
(2) *$\mathrm{Hom}(K_i[t], K_i) = 0$ for $i \geq 0$ and $t > 0$.*

Then there exists a cartesian object K of $D(\mathcal{C}_{total})$ whose associated simplicial system is isomorphic to (K_n, K_φ).

Proof. Set $X_n = g_{n!}K_n$ in $D(\mathcal{C}_{total})$. For each $n \geq 1$ we have

$$\mathrm{Hom}(X_n, X_{n-1}) = \mathrm{Hom}(K_n, g_n^{-1}g_{n-1!}K_{n-1}) = \bigoplus\nolimits_{\varphi:[n-1]\to[n]} \mathrm{Hom}(K_n, f_\varphi^{-1}K_{n-1})$$

Thus we get a map $X_n \to X_{n-1}$ corresponding to the alternating sum of the maps $K_\varphi^{-1} : K_n \to f_\varphi^{-1}K_{n-1}$ where φ runs over $\delta_0^n, \ldots, \delta_n^n$. We can do this because K_φ

is invertible by assumption (1). Please observe the similarity with the definition of the maps in the proof of Lemma 09WI. We obtain a complex

$$\ldots \to X_2 \to X_1 \to X_0$$

in $D(\mathcal{C}_{total})$. We omit the computation which shows that the compositions are zero. By Derived Categories, Lemma 0D83 if we have

$$\mathrm{Hom}(X_i[i-j-2], X_j) = 0 \text{ for } i > j+2$$

then we can extend this complex to a Postnikov system. The group is equal to

$$\mathrm{Hom}(K_i[i-j-2], g_i^{-1} g_{j!} K_j)$$

Again using that (K_n, K_φ) is cartesian we see that $g_i^{-1} g_{j!} K_j$ is isomorphic to a finite direct sum of copies of K_i. Hence the group vanishes by assumption (2). Let the Postnikov system be given by $Y_0 = X_0$ and distinguished sequences $Y_n \to X_n \to Y_{n-1} \to Y_n[1]$ for $n \geq 1$. We set

$$K = \mathrm{hocolim} Y_n[n]$$

To finish the proof we have to show that $g_m^{-1}K$ is isomorphic to K_m for all m compatible with the maps K_φ. Observe that

$$g_m^{-1}K = \mathrm{hocolim} g_m^{-1} Y_n[n]$$

and that $g_m^{-1}Y_n[n]$ is a Postnikov system for $g_m^{-1}X_n$. Consider the isomorphisms

$$g_m^{-1}X_n = \bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^{-1} K_n \xrightarrow{\bigoplus K_\varphi} \bigoplus\nolimits_{\varphi:[n]\to[m]} K_m$$

These maps define an isomorphism of complexes

$$\begin{array}{ccccccc}
\ldots & \longrightarrow & g_m^{-1}X_2 & \longrightarrow & g_m^{-1}X_1 & \longrightarrow & g_m^{-1}X_0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
\ldots & \longrightarrow & \bigoplus_{\varphi:[2]\to[m]} K_m & \longrightarrow & \bigoplus_{\varphi:[1]\to[m]} K_m & \longrightarrow & \bigoplus_{\varphi:[0]\to[m]} K_m
\end{array}$$

in $D(\mathcal{C}_m)$ where the arrows in the bottom row are as in the proof of Lemma 09WI. The squares commute by our choice of the arrows of the complex $\ldots \to X_2 \to X_1 \to X_0$; we omit the computation. The bottom row complex has a postnikov tower given by

$$Y'_{m,n} = \left(\bigoplus\nolimits_{\varphi:[n]\to[m]} \mathbf{Z} \to \ldots \to \bigoplus\nolimits_{\varphi:[0]\to[m]} \mathbf{Z} \right)[-n] \otimes^{\mathbf{L}}_{\mathbf{Z}} K_m$$

and $\mathrm{hocolim} Y'_{m,n} = K_m$ (please compare with the proof of Lemma 0D9I and Derived Categories, Example 0D8Z). Applying the second part of Derived Categories, Lemma 0D83 the vertical maps in the big diagram extend to an isomorphism of Postnikov systems provided we have

$$\mathrm{Hom}(g_m^{-1}X_i[i-j-1], \bigoplus\nolimits_{\varphi:[j]\to[m]} K_m) = 0 \text{ for } i > j+1$$

The is true if $\mathrm{Hom}(K_m[i-j-1], K_m) = 0$ for $i > j+1$ which holds by assumption (2). Choose an isomorphism given by $\gamma_{m,n} : g_m^{-1}Y_n \to Y'_{m,n}$ of Postnikov systems in $D(\mathcal{C}_m)$. By uniqueness of homotopy colimits, we can find an isomorphism

$$g_m^{-1}K = \mathrm{hocolim} g_m^{-1} Y_n[n] \xrightarrow{\gamma_m} \mathrm{hocolim} Y'_{m,n} = K_m$$

compatible with $\gamma_{m,n}$.

We still have to prove that the maps γ_m fit into commutative diagrams

$$\begin{array}{ccc} f_\varphi^{-1}g_m^{-1}K & \xrightarrow{K(\varphi)} & g_n^{-1}K \\ {\scriptstyle f_\varphi^{-1}\gamma_m}\downarrow & & \downarrow{\scriptstyle \gamma_n} \\ f_\varphi^{-1}K_m & \xrightarrow{K_\varphi} & K_n \end{array}$$

for every $\varphi : [m] \to [n]$. Consider the diagram

$$\begin{array}{ccccc} f_\varphi^{-1}(\bigoplus_{\psi:[0]\to[m]} f_\psi^{-1}K_0) & = & f_\varphi^{-1}g_m^{-1}X_0 & \xrightarrow[X_0(\varphi)]{} & g_n^{-1}X_0 & = & \bigoplus_{\chi:[0]\to[n]} f_\chi^{-1}K_0 \\ {\scriptstyle f_\varphi^{-1}(\bigoplus K_\psi)}\downarrow & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \bigoplus K_\chi} \\ f_\varphi^{-1}(\bigoplus_{\psi:[0]\to[m]} K_m) & & f_\varphi^{-1}g_m^{-1}K & \xrightarrow[K(\varphi)]{} & g_n^{-1}K & & \bigoplus_{\chi:[0]\to[n]} K_n \\ \| & & {\scriptstyle f_\varphi^{-1}\gamma_m}\downarrow & & \downarrow{\scriptstyle \gamma_n} & & \| \\ f_\varphi^{-1}Y'_{0,m} & \longrightarrow & f_\varphi^{-1}K_m & \xrightarrow{K_\varphi} & K_n & \longleftarrow & Y'_{0,n} \end{array}$$

The top middle square is commutative as $X_0 \to K$ is a morphism of simplicial objects. The left, resp. the right rectangles are commutative as γ_m, resp. γ_n is compatible with $\gamma_{0,m}$, resp. $\gamma_{0,n}$ which are the arrows $\bigoplus K_\psi$ and $\bigoplus K_\chi$ in the diagram. Going around the outer rectangle of the diagram is commutative as (K_n, K_φ) is a simplical system and the map $X_0(\varphi)$ is given by the obvious identifications $f_\varphi^{-1}f_\psi^{-1}K_0 = f_{\varphi\circ\psi}^{-1}K_0$. Note that the arrow $\bigoplus_\psi K_m \to Y'_{0,m} \to K_m$ induces an isomorphism on any of the direct summands (because of our explicit construction of the Postnikov systems $Y'_{i,j}$ above). Hence, if we take a direct summand of the upper left and corner, then this maps isomorphically to $f_\varphi^{-1}g_m^{-1}K$ as γ_m is an isomorphism. Working out what the above says, but looking only at this direct summand we conclude the lower middle square commutes as we well. This concludes the proof. $\square$